Greenhouse Gasses Emissions in World's Economy - Report

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Introduction

All data manipulations, visualizations, and insights were integrated and documented within an R Markdown file. R Markdown allowed me for seamless integration of R code, markdown an Latex languages to ensures reproducibility, clarity, and transparency in how the analysis was conducted. The report's source code can be found on my github website here.

In chart 1 the year to year percentage change was calculated for EU27, Euro area and World. The Euro area total GHG has to be summed GHG emission counties prior. I decided to present the data on a line chart with individual points highlighted - **Data Manipulations**: The raw data was processed using various R functions, including filtering, grouping, and summarizing, to deriving year to year percentage change and arithmetic mean. The `dplyr` and `tidyr` libraries were used to clean, transform, and aggregate the data where necessary. - **Plotting**: Visualizations were created using the `ggplot2` and `treemapify` package, which allowed for custom-designed plots to effectively communicate trends and relationships present in data.

Insights report

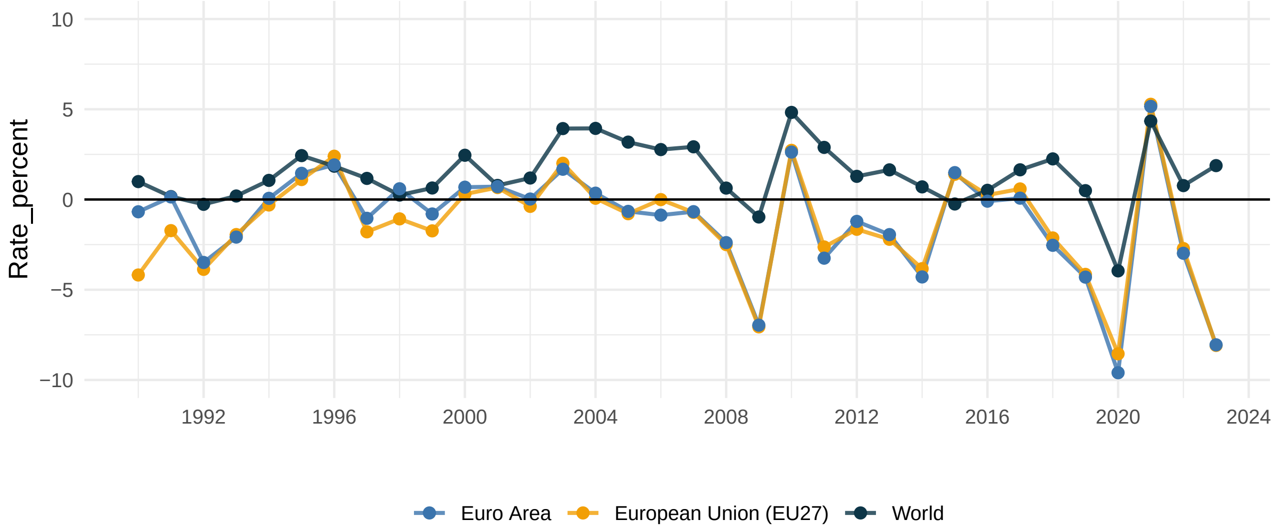
This report was prepared based on the EDGAR GHG Emissions database coming from 2024 'GHG emissions of all world countries' Publication (Crippa and Pekar 2024) (**report link**)

Evolution of GHG growth in the euro area, European Union (EU27) and worldwide

- In the European Union (EU27) and Euro Area, there was a noticeable 8.1% (or repetitively 261 Mt $CO_{2\text{-eq}}$ and 196 Mt $CO_{2\text{-eq}}$) decrease in total GHG emissions in 2023 compared to 2022. On the other hand, the World GHG emission increased by 1.8% (994 Mt $CO_{2\text{-eq}}$).
- From 1990 to 2023 the EU27 and Euro Area countries showed steady decline in GHG emissions with -1.2% compounded annual growth rate (CAGR) while Total World GHG emissions continue to raise with CAGR equal to 1.5%.

Greenhouse Gasses Emission − Year to year growth rate

% change of Total GHG emission



Comparison of countries' GHG emissions per capita aggregated according to the World Bank income groups;

- On average high-income countries emit around 11.8 Mt CO2 per capita whereas low-income countries emits almost 8 times less t Co2-eq with average of 1.5 Mt CO2 per capita which highlights disproportional emissions responsibility of wealthier nations.
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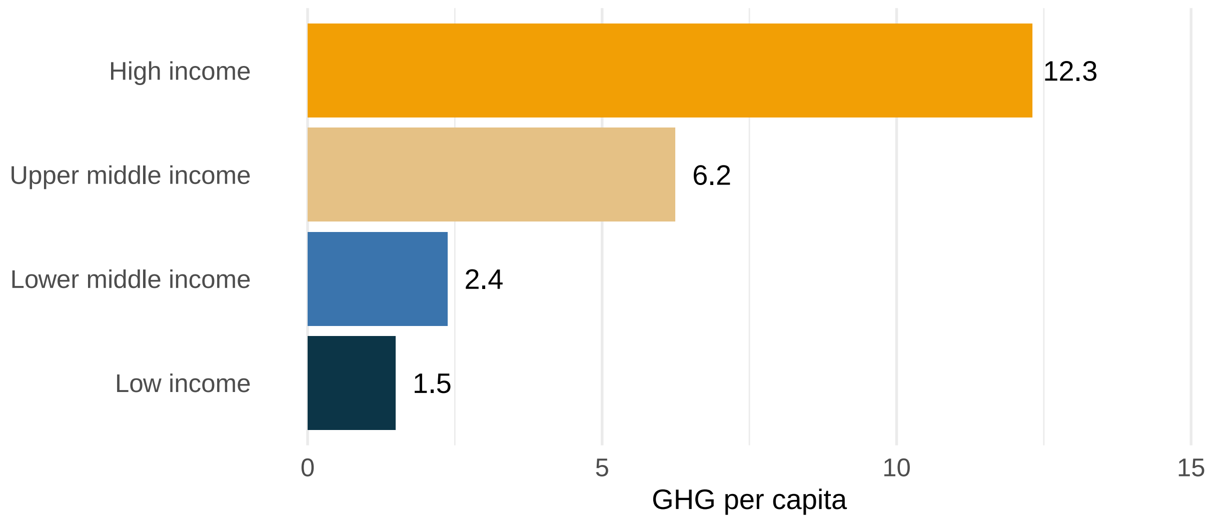
Greenhouse Gasses Emission per capita − Income groups comaparison in 2023

Tonnes of CO_2 – equivalent per person



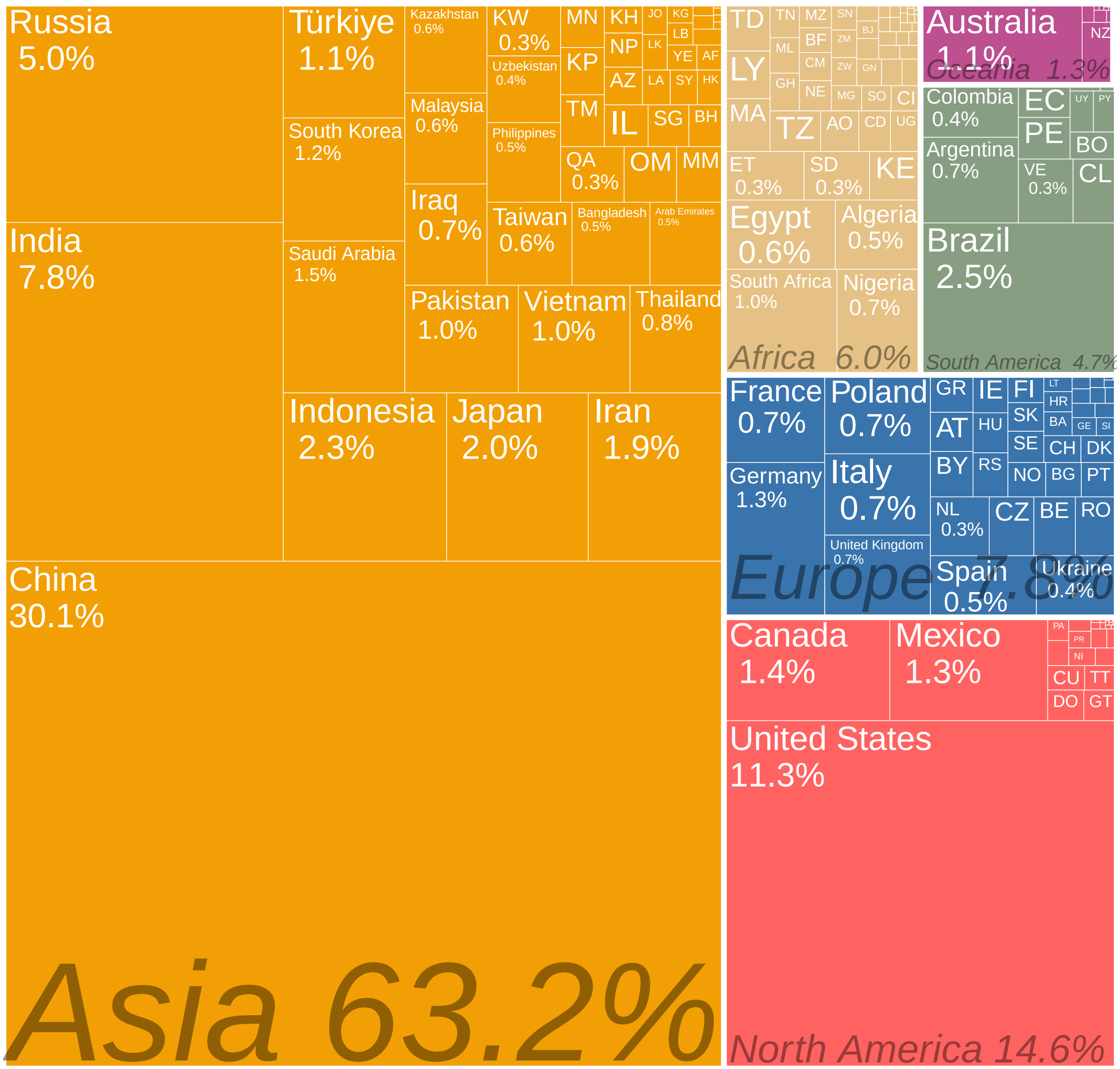
Note.(1) Individual countries were classified based on the latest clasification provided by World Bank (https://datatopics.worldbank.org/world−development−indicators/the−world−by−income−and−region.html)
(2) The World bank population total for each country were used to calculate the per capita indicator show on the chart (https://data.worldbank.org/indicator/SP.POP.T)

Contribution of individual country and continent GHG emissions to total world GHG emissions.

- China, the United States, India, Russia and Brazil were the world's largest GHG emitters in 2023. Together they account for 49.8% of global population and 62.7% of global GHG emissions.
- Asia is responsible for the largest share of emissions, mainly due to rapid industrialization in countries like China. The second
- In 2023, all EU27 countries except Croatia and Cyprus experienced a decrease in their emission levels compared to the previous year. In terms of contribution to the EU27's GHG emissions in 2023, Germany remained the largest emitter, followed by France, Italy, Poland and Spain.

Contribution of continent and individual country GHG emissions to total World GHG emissions

Proportion of Total GHG emission in %



References

Crippa, Guizzardi, M., and F. Pekar. 2024. "GHG Emissions of All World Countries." *Publications Office of the European Union, Luxembourg*. https://doi.org/10.2760/4002897.